Research Report
Network Encryption and Traffic Analysis

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Abstract

Your abstract.

1 Introduction

2 Background

2.1 The evolution of encryption

Encryption is the process of turning the original plain message into cipher text, an unrecognisable message different from the starting one. The reverse of encryption is decryption, and the entire process is called cryptography. In general cryptography focuses on privacy and the security of information based on data integrity, confidentiality and authentication. [1]
To secure the exchange of information over the Web, the HTTPS protocoll has been created to provide a cryptographic protection. HTTPS is based on other protection protocls that have been improved; it basically runs the HTTP protocol, which is at the center of the Web but does not provide confidentiality or integrity, on top of cryptography protocols, such as SSL and TLS, that allow bidirectional encryption in communication.
Because of the very different levels of security, it is believed that the HTTPS will replace mostly HTTP protocols, also thanks to initiatives and actions taken by the major browser vendors to promote the importance of security. [2]

3 Overview of the problem/evaluation

4 Examples

5 Discussion about mitigations/protections/applications

6 Conclusion

References

[1] K. John Singh and R. Manimegalai (2015), Evolution of Encryption Techniques and Data Security Mechanisms, World Applied Sciences Journal 33 (10): 1597-1613.

[2] S. Calzavara, R. Focardi, M. Nemec, A. Rabitti and M. Squarcina, "Postcards from the Post-HTTP World: Amplification of HTTPS Vulnerabilities in the Web Ecosystem," 2019 IEEE Symposium on Security and Privacy (SP), San Francisco, CA, USA, 2019, pp. 281-298.